KISSING CONFIGURATIONS OF 8358 POINTS IN EIGHTEEN DIMENSIONS

ABSTRACT. We describe several kissing configurations of 8358 points in $\mathbb{R}^{18}$. This ties the value 8358 reported in a public repository by Kravatskiy, with a different construction, and exceeds the best published lower bound $\tau_{18} \geq 7654$ of Cohn and Li by 704. All our configurations have the same body: the 7398 points of Ho's 19-dimensional kissing configuration that lie in the hyperplane orthogonal to $e_2 + e_{11} + e_{14}$. The remaining 960 points come from sign vectors of Ho's nonlinear code of minimum distance 5. Each has a 16-dimensional head and a tail in a two-dimensional plane, and the tails point in the six directions of a regular hexagon. The basic configuration projects these sign vectors into the hyperplane. We then prove that the tail of each added point can be rescaled independently by any factor in $[\sqrt{3}/2,\ 6 - 2\sqrt{6}]$. This one family contains the projection, a rational choice in which no added point touches any other point, a common rescaling that gives a non-antipodal spherical 3-design, and a two-color choice that pushes every cosine involving an added point below 0.4987532. From the last one we obtain an explicit nonlinear family of 8358-point configurations that fixes the body. We prove that four of these representatives are pairwise non-isometric. Exact computer checks confirm all pairs.

1. INTRODUCTION

The kissing problem in $\mathbb{R}^n$ asks for the largest set X of unit vectors with $\langle x, y\rangle \leq 1/2$ for all distinct $x, y \in X$ [3, Ch. 1, §2.2]. We write τ_n for this maximum. Cohn and Li improved the lower bounds in dimensions 17 through 21 [2]; in dimension 18 their bound is $\tau_{18} \geq 7654$ [2, Table 1.1]. Ho found a nonlinear code of 1280 words that raised the 19-dimensional bound to 11948 [4].

Theorem 1.1. *There are kissing configurations of* 8358 *points in* $\mathbb{R}^{18}$. *In particular* $\tau_{18} \geq 8358$.

Table 1 puts this value in context. The public repository of Kravatskiy [5] reported 8358 points in dimension 18 first. It was obtained by a different construction. We have not tested whether that configuration is isometric to any of ours. Our value is therefore a tie with that repository value, and an improvement of 704 over the best published value. We make no priority claim.

TABLE 1. Kissing number bounds in dimension 18, as of October 2026 [1]. The upper bound is the floor of a numerical semidefinite programming value [6] and is not interval-certified.

dimension	published lower bound	outside repository	this paper	upper bound
18	7654 [2]	8358 [5]	8358	16406 [6]

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The construction in outline. All our configurations share the same 7398 points, which we call the *body*. It arises in three steps.

(1) Start with Ho's configuration in $\mathbb{R}^{19}$. It consists of the Cohn-Li base of 10668 vectors of squared norm 8 and 1280 scaled sign vectors (Section 2).
(2) Cut the base by the hyperplane $s^\perp$, where $s = e_2 + e_{11} + e_{14}$. The coordinate set $\{2, 11, 14\}$ is the only word of weight 3 in the linear code D spanned by Ho's generators. Hence every octad meets it in 0 or 2 places, and exactly 7398 base vectors lie in $s^\perp$. These form the body.
(3) Of the 1280 sign vectors q, exactly 960 have $\langle q, s\rangle = \pm 1$. Their projections $3q - \langle q, s\rangle\, s$ into $s^\perp$ are compatible with the body and with each other.

This gives $7398 + 960 = 8358$ (Section 3).

Each of the 960 added points splits into a 16-dimensional *head* and a *tail* in the plane $s^\perp \cap \mathbb{R}^{\{2,11,14\}}$. The tails point in the six directions of a regular hexagon. In Section 4 we prove that the tail of each added point can be multiplied by its own factor $\lambda \in [\sqrt{3}/2,\, 6 - 2\sqrt{6}]$. Every member of this family has 8358 points. Section 5 describes four members and a family of nonlinear motions:

- the projection itself ($\lambda = 1$), with integer coordinates;
- an *irregular* rational choice, in which no added point touches any other point;
- a common factor $\lambda^2 = 174/151$, which gives a spherical 3-design with no antipodal pairs among the added points;
- a *two-color* choice with one factor for each class of the hexagon bipartition, in which every cosine involving an added point is at most 1246883/2500000;
- a family of nonlinear motions of the two-color configuration that fixes the body and moves the added points.

We show that the first four are pairwise non-isometric (Section 6). Section 7 lists what was proved by hand and what by exact computation.

Notation. Coordinates of $\mathbb{R}^{19}$ are numbered $1, \dots, 19$, and

$$T = \{2, 11, 14\}, \qquad H = \{1, \dots, 19\} \setminus T, \qquad s = \mathbf{1}_T = e_2 + e_{11} + e_{14}.$$

We call H the *head* and T the *tail*. For $x \in \mathbb{R}^{19}$ we write x_H and x_T for the vectors that agree with x on H and on T and vanish elsewhere, so $x = x_H + x_T$. The hyperplane $s^\perp$ is our copy of $\mathbb{R}^{18}$. Inner products are those of $\mathbb{R}^{19}$ restricted to $s^\perp$.

For nonzero x, y the angle between them is at least 60° if and only if

$$\langle x, y\rangle \le 0 \quad \text{or} \quad 4\langle x, y\rangle^2 \le |x|^2 |y|^2. \tag{1}$$

We call such a pair *compatible*. If every pair from a finite set of nonzero vectors is compatible, the normalized vectors form a kissing configuration. Distinct vectors give distinct points, because (1) excludes cosine 1. A pair with $\langle x, y\rangle = \frac{1}{2}|x||y|$ is a *contact*. We often check compatibility in the form $\langle x, y\rangle \le \frac{1}{2}|x|\,|y|$ and call the right side the *threshold* of the pair.

2. Ho's configuration

We identify a binary word with its support. For $c \in \mathbb{F}_2^{19}$ we write $q(c) = (-1)^c \in \{\pm 1\}^{19}$ for its sign vector. If c, c' are at Hamming distance d, then $\langle q(c), q(c')\rangle = 19 - 2d$.

2.1. **The codes.** Let $D \le \mathbb{F}_2^{19}$ be the linear code spanned by Ho's twelve words [4, §2]

$$\begin{aligned} m_1 &= \{1,8,9,12,16,17,18,19\}, & m_2 &= \{2,10,11,14,15,17,18\},\\ m_3 &= \{3,7,9,13,15,16,17,19\}, & m_4 &= \{4,7,8,10,12,15,16,19\},\\ m_5 &= \{5,10,12,13,15,16,17,18\}, & m_6 &= \{6,7,8,9,10,13,16,18\},\\ s_1 &= \{1,4,7,9\}, & s_2 &= \{1,5,6,18\},\\ s_3 &= \{1,3,12,15\}, & s_4 &= \{1,10,13,19\},\\ r_1 &= \{1,3,5,6,7,13,14,15,18\}, & r_2 &= \{2,4,6,7,8,13,14,16,17,18\}. \end{aligned}$$

Let $M = \langle m_1, \dots, m_6 \rangle$, let $s_5 = \{3,5,7,10\} = s_1 + s_2 + s_3 + s_4 + m_4 + m_6$, and let $R = \{0, r_1, r_2, r_1 + r_2\}$. Ho's code is

$$A = \bigcup_{k=1}^{5} \bigcup_{r \in R} (M + s_k + r) \ \subseteq D.$$

This is the code of [4, Proposition 9, §4]. Ho proves that D is a punctured extended Golay code [4, Proposition 2], and that $|A| = 1280$ and A has minimum distance 5 [4, Proposition 9].

Lemma 2.1. *The following hold.*

(1) $\dim D = 12$, *and* $T = \{2,11,14\}$ *is the only word of* D *of weight* 3.
(2) *The dual code* $D^\perp$ *has exactly* 78 *words of weight* 8, *which we call octads. Two distinct octads meet in* 0, 2 *or* 4 *coordinates.*
(3) *Every octad meets* T *in* 0 *or* 2 *coordinates. Exactly* 30 *octads are disjoint from* T, *and for each* 2*-subset of* T *exactly* 16 *octads meet* T *in that subset.*
(4) $|A| = 1280$, $A \subseteq D$, *and distinct words of* A *are at Hamming distance at least* 5.

Proof. The first claim in (3) needs no computer. Every octad lies in $D^\perp$ and $T \in D$, so each octad meets T in an even number of coordinates, and $|T| = 3$.

The rest is a finite enumeration of the 4096 words of D and the 128 words of $D^\perp$, in exact arithmetic. The weight distribution of D is

$$1, 1, 20, 120, 280, 308, 438, 880, 880, 438, 308, 280, 120, 20, 1, 1 \quad \text{in weights } 0, 3, 4, \dots, 16, 19.$$

The dual $D^\perp$ has 78 words of weight 8, 48 of weight 12 and one of weight 16. Part (4) is Ho's [4, Proposition 9]; we also checked all $\binom{1280}{2}$ distances. □

2.2. **The base and the parity lemma.** Let B be the set of the $4\binom{19}{2} = 684$ *roots* $\pm 2e_i \pm 2e_j$ together with, for each octad o, the 128 vectors with entries ± 1 on o, zeros elsewhere and an odd number of minus signs. These $78 \cdot 128 = 9984$ vectors are the *octad vectors*; the 128 on o form the *fiber* over o. All 10668 vectors of B have squared norm 8. This is the odd-sign configuration of Cohn and Li [2, §2]. Ho adds to it the scaled sign vectors $\sqrt{8/19}\, q(c)$, $c \in A$.

Lemma 2.2 (Cohn and Li)**.** *Distinct* $b, b' \in B$ *satisfy* $\langle b, b' \rangle \le 4$.

Proof. Two distinct roots have inner product in $\{0, \pm 4, -8\}$. A root has two nonzero coordinates, each contributing at most 2 against an octad vector. Two vectors on the same octad have odd sign patterns, so they differ in an even positive number d of places and have inner product $8 - 2d \le 4$. Two vectors on different octads share at most four coordinates by Lemma 2.1(2). □

Lemma 2.3 (parity)**.** *Let* $c \in D$ *and* $q = q(c)$. *For an octad vector* b *we have* $\langle b, q \rangle \le 6$. *For a root* b *we have* $\langle b, q \rangle \le 4$.

Proof. Let b lie on the octad o. Since $o \in D^\perp$, the word c meets o in an even number of coordinates. So q has an even number of minus signs on o, while b has an odd number. Hence q and b disagree in an odd number $m \geq 1$ of coordinates of o, and $\langle b, q\rangle = 8 - 2m \leq 6$. A root has two entries ± 2, so $\langle b, q\rangle \leq 4$. □

3. The section and the projection

3.1. **The body.** Let $B_s = \{b \in B : \langle b, s\rangle = 0\}$. We sort its vectors into four types, which recur throughout the paper: head roots, tail roots, full fibers and half fibers.

Lemma 3.1. $|B_s| = 486 + 3840 + 3072 = 7398$, *and* B_s *spans* $s^\perp$.

Proof. Head roots. A root $\pm 2e_i \pm 2e_j$ with $i, j \in H$ lies in $s^\perp$; there are $4\binom{16}{2} = 480$.

Tail roots. A root with exactly one index in T has $\langle b, s\rangle = \pm 2$, so it is not in B_s. A root with both indices in T lies in $s^\perp$ exactly when its two signs are opposite. This gives the 6 roots $2(e_i - e_j)$, $i \neq j \in T$. In total there are $480 + 6 = 486$ roots.

Full fibers. If an octad is disjoint from T, its whole fiber lies in $s^\perp$. By Lemma 2.1(3) these are $30 \cdot 128 = 3840$ vectors.

Half fibers. Let $o \cap T = \{i, j\}$. An octad vector b on o has $\langle b, s\rangle = b_i + b_j$, which vanishes exactly when $b_j = -b_i$. Then b_i, b_j contribute exactly one minus sign, so the six entries on $o \setminus T$ must have an even number of minus signs. This gives $2 \cdot 2^5 = 64$ vectors, half the fiber. By Lemma 2.1(3) these are $48 \cdot 64 = 3072$ vectors.

Span. The head roots span $\mathbb{R}^H$, and the tail roots span the plane $\{x \in \mathbb{R}^T : x_2 + x_{11} + x_{14} = 0\}$. Together they span $s^\perp$. □

3.2. **The kept words.** For $c \in \mathbb{F}_2^{19}$ put $q = q(c)$ and $h(c) = \langle q, s\rangle = q_2 + q_{11} + q_{14} \in \{\pm 1, \pm 3\}$.

Lemma 3.2. *Let* $A_1 = \{c \in A : h(c) = \pm 1\}$. *Then*

$$A_1 = \bigcup_{k=1}^{5} \bigcup_{r \in \{r_1, r_2, r_1 + r_2\}} (M + s_k + r), \qquad |A_1| = 960.$$

The restriction $c \cap T$ *of a word* $c \in A_1$ *takes each of the six values* $\{14\}$, $\{2, 11\}$, $\{2, 14\}$, $\{11\}$, $\{2\}$, $\{11, 14\}$ *for exactly* 160 *words. The* 480 *words with a one-element restriction have* $h = 1$*; the* 480 *with a two-element restriction have* $h = -1$.

Proof. We read off the generators. Among $m_1, \ldots, m_6$ only m_2 meets T, and $m_2 \supseteq T$; so the restrictions to T of words of M are exactly $\emptyset$ and T, each for 32 words. None of $s_1, \ldots, s_5$ meets T. Finally $r_1 \cap T = \{14\}$, $r_2 \cap T = \{2, 14\}$ and $(r_1 + r_2) \cap T = \{2\}$. Hence a word of $M + s_k + r$ has restriction $r \cap T$ or $(r \cap T) + T$, each for 32 words.

For $r = 0$ the restriction is $\emptyset$ or T, so $h = \pm 3$ and the word is not in A_1. For $r \neq 0$ the restriction has one or two elements, so $h = \pm 1$. This gives $5 \cdot 3 \cdot 64 = 960$ words, 160 for each of the six restrictions. A one-element restriction has one minus sign among q_2, q_{11}, q_{14}, so $h = 1$; a two-element restriction has $h = -1$. □

3.3. **Tails and heads.** For $c \in A_1$ the *tail vector* is

$$\tau(c) = 3q_T - h(c)\, s.$$

It is the orthogonal projection of $3q_T$ onto the plane $s^\perp \cap \mathbb{R}^T$.

Lemma 3.3 (tails)**.** *Let* $c, c' \in A_1$.

(1) $\tau(c)$ *lies in the plane* $s^\perp \cap \mathbb{R}^T$ *and* $|\tau(c)|^2 = 24$. *Up to permutation of the three tail coordinates,* $\tau(c) = \pm(2, 2, -4)$.

(2) *The six values of* τ *are the vertices of a regular hexagon, each taken by* 160 *words.*
(3) $\langle \tau(c), \tau(c') \rangle$ *is* 24, 12, -12 *or* -24 *according as the restrictions* $c \cap T$ *and* $c' \cap T$ *differ in* 0, 1, 2 *or* 3 *tail coordinates. We call tails with inner product* 12 adjacent.
(4) *Adjacent tails have opposite values of* h. *So the two triangles* $h = 1$ *and* $h = -1$ *are the two classes of the bipartition of the hexagon.*

Proof. (1) $\langle \tau(c), s \rangle = 3h - 3h = 0$, and $|\tau(c)|^2 = 9 \cdot 3 - 6h^2 + 3h^2 = 24$ since $h^2 = 1$. If $c \cap T = \{14\}$ then $q_T = (1, 1, -1)$ on $(2, 11, 14)$, $h = 1$ and $\tau = (2, 2, -4)$; the other five cases are permutations and negations of this one.

(2) follows from (1) and Lemma 3.2. For (3), $\langle \tau, \tau' \rangle = 9\langle q_T, q'_T \rangle - 3hh'$, and $\langle q_T, q'_T \rangle = 3 - 2e$ where e is the number of tail coordinates in which the restrictions differ. If $e \in \{0, 2\}$ the two restrictions have the same size parity, so $hh' = 1$ and the product is $27 - 18e - 3 \in \{24, -12\}$. If $e \in \{1, 3\}$ then $hh' = -1$ and the product is $27 - 18e + 3 \in \{12, -24\}$. (4) is the case $e = 1$. □

Lemma 3.4 (head distances, computed)**.** *Let* $c \neq c'$ *in* A_1, *with* d_H *the number of head coordinates in which they differ.*

(1) *The* 960 *heads* $q(c)_H$ *are distinct.*
(2) *If* $\tau(c) = \tau(c')$, *then* $d_H \geq 6$, *so* $\langle q(c)_H, q(c')_H \rangle \leq 4$.
(3) *If* $\tau(c) \neq \tau(c')$, *then* $d_H \geq 4$, *so* $\langle q(c)_H, q(c')_H \rangle \leq 8$.

Exactly 13056 *unordered pairs have adjacent tails and* $d_H = 4$.

Proof. Direct check of the $\binom{960}{2} = 460320$ pairs in exact integers. The minimum head distance is 6 for equal tails and 4 for each of the three other tail relations. □

Minimum distance 5 alone gives only $d_H \geq 5 - e \geq 2$ for tails that differ in e coordinates. Lemma 3.4 is stronger whenever $e \neq 1$, and this extra room is what Section 4 uses.

3.4. **The projection.** For $c \in A_1$ let

$$y(c) = 3q(c) - h(c)\, s = 3q(c)_H + \tau(c).$$

Then $\langle y(c), s \rangle = 0$ and $|y(c)|^2 = 9 \cdot 16 + 24 = 168$.

Theorem 3.5. *The* 7398 *vectors of* B_s *and the* 960 *vectors* $y(c)$, $c \in A_1$, *are pairwise compatible. Their normalizations form a kissing configuration of* 8358 *points in* $s^\perp \cong \mathbb{R}^{18}$, *spanning* $s^\perp$.

Proof. There are three kinds of pair.

Body and body. By Lemma 2.2, $\langle b, b' \rangle \leq 4 = \frac{1}{2} \cdot 8$.

Body and added. For $b \in B_s$ we have $\langle b, s \rangle = 0$, so $\langle b, y(c) \rangle = 3\langle b, q(c) \rangle \leq 18$ by Lemma 2.3. Since $4 \cdot 18^2 = 1296 < 1344 = 8 \cdot 168$, the pair is compatible, with room to spare.

Added and added. If $c, c' \in A_1$ are at Hamming distance d, then, using $\langle q(c'), s \rangle = h(c')$ and $|s|^2 = 3$,

$$\langle y(c), y(c') \rangle = 9(19 - 2d) - 3h(c)h(c') - 3h(c')h(c) + 3h(c)h(c') = 9(19 - 2d) - 3h(c)h(c').$$

By Lemma 2.1(4), $d \geq 5$, so this is at most $81 + 3 = 84 = \frac{1}{2} \cdot 168$.

The span claim is Lemma 3.1, and the count is Lemmas 3.1 and 3.2. □

Remark 3.6. The same proof applies to the hyperplane orthogonal to $\varepsilon_2 e_2 + \varepsilon_{11} e_{11} + \varepsilon_{14} e_{14}$ for any signs ε_i, with $h(c) = \sum_{i \in T} \varepsilon_i q_i$. The norms, the formula for $\langle y(c), y(c') \rangle$ and the bound $\langle b, y(c) \rangle = 3\langle b, q \rangle$ are unchanged. A count by computer gives $7398 + 960 = 8358$ points for each of the four sign classes. We have not determined whether these four configurations are isometric, and we do not use them below.

4. Independent tail factors

Write each added vector as head plus tail, $y(c) = 3q(c)_H + \tau(c)$. We now multiply the tail by a factor that may depend on c.

Theorem 4.1. *For each $c \in A_1$ choose λ_c with*

$$\frac{\sqrt{3}}{2} \;\le\; \lambda_c \;\le\; 6 - 2\sqrt{6},$$

and put $p(c) = 3q(c)_H + \lambda_c \tau(c) \in s^\perp$. Then the 7398 vectors of B_s and the 960 vectors $p(c)$ are pairwise compatible, so their normalizations form a kissing configuration of 8358 points in $s^\perp$. A contact involving some $p(c)$ occurs only in the following cases:

1. *$\lambda_c = \sqrt{3}/2$, against an octad vector b with $\langle b_H, q(c)_H\rangle = 6$ and $\langle b_T, q(c)_T\rangle = 0$;*
2. *$\lambda_c = 6 - 2\sqrt{6}$, against a half-fiber octad vector;*
3. *against $p(c')$, where $\tau(c), \tau(c')$ are adjacent, $d_H(c, c') = 4$ and $\lambda_c = \lambda_{c'}$.*

The proof has two halves. For a body vector b against $p(c)$, the inner product splits as $3(\alpha + \lambda\delta)$ with α a head product and δ a tail product. The four body types of Lemma 3.1 give four one-line inequalities in λ (Table 2). The lower end $\sqrt{3}/2$ comes from full fibers, the upper end $6 - 2\sqrt{6}$ from half fibers. For two added vectors, Lemma 3.4 and the Cauchy-Schwarz inequality finish the job.

Proof. Throughout, $q = q(c)$, $\tau = \tau(c)$ and $\lambda = \lambda_c$. We have $|p(c)|^2 = 9\cdot 16 + 24\lambda^2 = 24(6+\lambda^2)$. Note that $\sqrt{3}/2 \approx 0.866$ and $6 - 2\sqrt{6} \approx 1.101$, so $\frac{3}{4} \le \lambda^2 < 3$.

Body and added. Let $b \in B_s$. The threshold is $\frac{1}{2}|b|\,|p(c)| = \sqrt{48(6+\lambda^2)}$. Put

$$\alpha = \langle b_H, q_H\rangle, \qquad \delta = \langle b_T, q_T\rangle.$$

Since $\langle b_T, s\rangle = \langle b, s\rangle = 0$, we have $\langle b_T, \tau\rangle = 3\langle b_T, q_T\rangle - h\langle b_T, s\rangle = 3\delta$, and so

$$\langle b, p(c)\rangle = 3(\alpha + \lambda\delta). \tag{2}$$

We bound α and δ for each type of b.

Table 2. The four body types against $p(c)$. The last column is the condition $\langle b, p(c)\rangle \le \sqrt{48(6+\lambda^2)}$ for the bound in the fourth column.

type of b	δ	bound on α	bound on $\langle b, p(c)\rangle$	condition on λ
head root	0	4	12	always, since $144 < 288$
tail root	≤ 4	0	12λ	$\lambda^2 \le 3$
full fiber	0	6	18	$\lambda^2 \ge 3/4$
half fiber, $\delta = 0$	0	6	18	$\lambda^2 \ge 3/4$
half fiber, $\delta = -2$	-2	6	$18 - 6\lambda$	always
half fiber, $\delta = 2$	2	4	$12 + 6\lambda$	$\lambda^2 - 12\lambda + 12 \ge 0$

Head roots. $b_T = 0$ and $\alpha \le 4$ by Lemma 2.3. So $\langle b, p(c)\rangle \le 12 < \sqrt{288} \le \sqrt{48(6+\lambda^2)}$.

Tail roots. $b = 2\varepsilon(e_i - e_j)$ with $i, j \in T$, so $\alpha = 0$ and $\delta = 2\varepsilon(q_i - q_j) \le 4$. Hence $\langle b, p(c)\rangle \le 12\lambda$, and $144\lambda^2 \le 48(6+\lambda^2)$ is equivalent to $\lambda^2 \le 3$.

Full fibers. $b_T = 0$ and $\alpha = \langle b, q\rangle \le 6$ by Lemma 2.3. So $\langle b, p(c)\rangle \le 18$, and $324 \le 48(6+\lambda^2)$ is equivalent to $\lambda^2 \ge 3/4$, with equality only at $\lambda = \sqrt{3}/2$.

Half fibers. Let b lie on an octad o with $o \cap T = \{i, j\}$. Then $b_T = \varepsilon(e_i - e_j)$, so $\delta = \varepsilon(q_i - q_j) \in \{0, \pm 2\}$. Two facts bound α: Lemma 2.3 gives $\alpha + \delta = \langle b, q\rangle \le 6$, and $\alpha \le 6$

because b_H has six nonzero entries. If $\delta = 0$, then $\langle b, p(c)\rangle \le 18$ as for full fibers. If $\delta = -2$, then $\langle b, p(c)\rangle \le 3(6 - 2\lambda) < 18$. If $\delta = 2$, then $\alpha \le 4$ and $\langle b, p(c)\rangle \le 12 + 6\lambda$. Now

$$(12 + 6\lambda)^2 \le 48(6 + \lambda^2) \iff 12\lambda^2 - 144\lambda + 144 \ge 0 \iff \lambda^2 - 12\lambda + 12 \ge 0,$$

which holds for $0 < \lambda \le 6 - 2\sqrt{6}$, with equality only at $\lambda = 6 - 2\sqrt{6}$.

Added and added. Let $c \ne c'$ with factors λ, λ', and write $q' = q(c')$, $\tau' = \tau(c')$. Then

$$\langle p(c), p(c')\rangle = 9\langle q_H, q'_H\rangle + \lambda\lambda'\langle \tau, \tau'\rangle,$$

and the threshold is $\frac{1}{2}|p(c)|\,|p(c')| = 12\sqrt{(6 + \lambda^2)(6 + \lambda'^2)} > 72$. By Lemma 3.3(3) there are three cases for $\langle \tau, \tau'\rangle$.

- Equal tails, $\langle \tau, \tau'\rangle = 24$. Lemma 3.4(2) gives $\langle p(c), p(c')\rangle \le 36 + 24\lambda\lambda' \le 36 + 24(6 - 2\sqrt{6})^2 < 66 < 72$.
- Adjacent tails, $\langle \tau, \tau'\rangle = 12$. Lemma 3.4(3) gives

  $$\langle p(c), p(c')\rangle \le 72 + 12\lambda\lambda' = 12\big(\sqrt{6}\cdot\sqrt{6} + \lambda\lambda'\big) \le 12\sqrt{(6 + \lambda^2)(6 + \lambda'^2)}$$

  by the Cauchy-Schwarz inequality for the vectors $(\sqrt{6}, \lambda)$ and $(\sqrt{6}, \lambda')$. Equality needs $\lambda = \lambda'$ and $d_H = 4$.
- Otherwise $\langle \tau, \tau'\rangle < 0$, so $\langle p(c), p(c')\rangle < 9 \cdot 8 = 72$.

Body pairs are covered by Lemma 2.2. The equality cases found above are exactly the three listed in the theorem. □

Corollary 4.2. *If every λ_c lies in the open interval $(\sqrt{3}/2,\ 6 - 2\sqrt{6})$ and $\lambda_c \ne \lambda_{c'}$ whenever $\tau(c), \tau(c')$ are adjacent and $d_H(c, c') = 4$, then no added point touches any other point. The only contacts are the contacts of the body.*

In terms of the squared tail radius t of the scaled point $q_H + \frac{\lambda}{3}\tau$, whose head is a ± 1 vector, $t = 8\lambda^2/3$, and Theorem 4.1 allows $t \in [2,\ 160 - 64\sqrt{6}] \approx [2,\ 3.2327]$.

5. Representatives

5.1. **The projection.** The choice $\lambda_c = 1$ for all c gives Theorem 3.5. Its points are integer vectors: 7398 of squared norm 8 and 960 of squared norm 168. By Theorem 4.1(3) its contacts that involve an added point are exactly the 13056 pairs of Lemma 3.4. In all it has 1 367 430 contacts, of which 1 354 374 lie in the body (computer count).

5.2. **An irregular rational choice.** Choose 960 distinct rational factors λ_c in the interval $[301/300,\ 147/142]$. Both ends lie strictly inside $(\sqrt{3}/2,\ 6 - 2\sqrt{6})$, so Corollary 4.2 applies: no added point touches anything. Scaling each $p(c)$ by a common denominator gives integer vectors. The irregular representative is defined by the explicit list of factors in the accompanying data; it uses the endpoints 301/300 and 147/142. For this choice, every pair involving an added point has cosine at most

$$\frac{1}{2} - \frac{1}{40\,072\,218}$$

(exact computer check). The contacts are the 1 354 374 contacts of the body.

This slack gives independent motions. Move each added unit point by any amount at most $\eta = 1/80\,144\,436$ within the unit sphere of $s^\perp$ and keep the body fixed. Every inner product involving an added point changes by at most $2\eta = 1/40\,072\,218$, so the result is again a kissing configuration of 8358 points. This is an open family of dimension $960 \cdot 17 = 16320$ around the irregular configuration. These motions never add a point.

5.3. **A non-antipodal spherical 3-design.** Recall that a finite set X of unit vectors in $\mathbb{R}^{18}$ is a spherical 3-design when

$$\sum_{x\in X} x = 0, \qquad \sum_{x\in X} xx^{\mathsf{T}} = \frac{|X|}{18}\, I_{18}, \qquad \sum_{x\in X} x\otimes x\otimes x = 0.$$

Proposition 5.1. *Take $\lambda_c = \beta$ for all c, with $\beta^2 = 174/151$. The resulting 8358 unit vectors in $s^\perp$ form a spherical 3-design. The body is antipodal, and none of the 960 added points has its antipode in the configuration.*

The proof checks the three moments in turn. The body is antipodal, so it kills the odd moments by itself; the odd moments of the added points vanish by computer. For the second moment we split the 18×18 matrix into head, tail and mixed blocks. The head and mixed blocks are computer checks. The tail block is a short hand computation, and it is where the value of β comes from.

Proof. Since $\sqrt{3}/2 < \beta \approx 1.0735 < 6-2\sqrt{6}$, Theorem 4.1 applies. The added vectors have common squared norm $144 + 24\beta^2 = 25920/151$.

Odd moments. The body is antipodal: negating a root gives a root, and negating an octad vector changes eight signs and keeps the parity. So the body contributes nothing to the first and third moments. For the added points, each entry of $\sum p\otimes p\otimes p$ is an integer multiple of β^k, where k is the number of tail indices among the three positions. We checked by computer that all these integers vanish, and similarly for the first moment. (The first moment also vanishes by hand: every coordinate lies in the support of some m_i, so $\sum_{m\in M}(-1)^{(m+v)_j} = 0$ for all v and j.)

Second moment, head and mixed blocks. The head block equals $\frac{8358}{18} I_H$ by computer. The mixed head-tail block of the added points is an integer matrix times β and vanishes by computer; so does the mixed block of the body.

Second moment, tail block. Let P_T be the orthogonal projection of $\mathbb{R}^{19}$ onto the plane $s^\perp\cap\mathbb{R}^T$. We use that a regular m-gon of vectors v of squared radius ρ in a plane has $\sum vv^{\mathsf{T}} = \frac{m\rho}{2}$ times the projection onto that plane. The tail block has three contributions.

- The six tail roots $2(e_i - e_j)$ form a regular hexagon of squared radius 8, giving $24P_T$.
- Each half-fiber vector has $b_T = \pm(e_i - e_j)$. For each pair $\{i,j\}\subseteq T$ there are $16\cdot 64 = 1024$ of them, 512 with each sign. So each of the six vectors $\pm(e_i-e_j)$, $i\neq j\in T$, occurs 512 times, and these six form a regular hexagon of squared radius 2. This gives $512\cdot 6\, P_T = 3072P_T$.
- The 960 tails $\tau(c)$ form 160 copies of a regular hexagon of squared radius 24, giving $160\cdot 72\, P_T = 11520P_T$ before scaling by β^2.

Body vectors have squared norm 8. So the tail block of $\sum xx^{\mathsf{T}}$ is

$$\frac{24+3072}{8}P_T + \frac{151\,\beta^2}{25920}\cdot 11520\, P_T = \Big(387 + \frac{1392}{18}\Big)P_T = \frac{8358}{18}\, P_T,$$

using $\beta^2 = 174/151$. Hence $\sum xx^{\mathsf{T}} = \frac{8358}{18}$ times the identity of $s^\perp$.

Antipodes. $-p(c)$ is not a body vector, since its head has sixteen nonzero entries. It is not an added vector either: $-p(c) = p(c')$ would force $q(c') = -q(c)$, and no two words of A_1 are complements of each other (computer check). □

The design has the same number of contacts as the projection, $1\,354\,374 + 13\,056 = 1\,367\,430$, since all its added points share one factor. The cosines between body and added points have the form $(u+v\beta)/\sqrt{8\cdot 25920/151}$ with integers u, v. Some half-fiber pairs with $\delta = 2$ have

$uv \neq 0$, for example those with $\alpha = 4$, where $u = 12$ and $v = 6$. Their squared cosines are irrational, so no orthonormal basis gives the points rational coordinates, even after scaling each point. We checked all pairs of the design exactly in $\mathbb{Q}(\beta)$. Its largest cosine below $\frac{1}{2}$ is $g(\beta) \approx 0.49763$, with g as in the next subsection.

5.4. **A two-color choice.** Give the factor a to the 480 words with $h = 1$ and the factor b to the 480 words with $h = -1$. Adjacent tails have opposite h by Lemma 3.3(4), so $a \neq b$ makes every added point contact-free by Corollary 4.2. Let

$$f(\lambda) = \frac{18}{\sqrt{192(6+\lambda^2)}}, \qquad g(\lambda) = \frac{12+6\lambda}{\sqrt{192(6+\lambda^2)}}, \qquad k(a,b) = \frac{6+ab}{2\sqrt{(6+a^2)(6+b^2)}}.$$

These are the cosines of the three tight cases in the proof of Theorem 4.1: a full fiber (or a half fiber with $\delta = 0$) with $\alpha = 6$, a half fiber with $\delta = 2$ and $\alpha = 4$, and two adjacent added points with $d_H = 4$. Here $192(6+\lambda^2) = |b|^2|p(c)|^2$.

Proposition 5.2. *Let $a, b \in [\sqrt{3}/2,\ 6-2\sqrt{6}]$ with $a \leq b$. In the two-color configuration, the largest cosine of a pair involving an added point is*

$$\max\{f(a),\ g(b),\ k(a,b)\}.$$

For $a = 22133/25000$ and $b = 27161/25000$ it equals $f(a) = \sqrt{1054687500/4239869689} \approx 0.4987531$, which is at most $1246883/2500000$. For every choice of $a \leq b$ in the interval it exceeds 0.4987528. The same holds for $a > b$, with the roles of a and b exchanged.

The proof has three steps: the maximum is at most the stated value, each of the three values is attained, and no choice of a, b pushes all three below 0.4987528.

Proof. Upper bound. The proof of Theorem 4.1 bounds each cosine involving an added point. On the interval, f is decreasing and g is increasing (its derivative has the sign of $6 - 2\lambda$). The remaining cases (roots, half fibers with $\delta = -2$, equal or non-adjacent tails) have cosine below 0.39, while $f(\lambda) \geq f(6-2\sqrt{6}) > 0.48$. So the largest cosine is at most

$$\max\{f(a), f(b), g(a), g(b), k(a,b)\} = \max\{f(a), g(b), k(a,b)\}.$$

The three values are attained. For f: take any added word c of the color with factor a, an octad o disjoint from T, and the octad vector on o that agrees with $q(c)$ except in one coordinate. It has odd parity by the proof of Lemma 2.3, and $\alpha = 6$. For g: take c of the color with factor b. Some $i, j \in T$ have $q_i \neq q_j$. Take one of the 16 octads o with $o \cap T = \{i, j\}$, and the octad vector b on o with $b_i = q_i$, $b_j = q_j$ and b_H equal to q_H on o except in one coordinate. Then b has one disagreement with q, so odd parity, $\langle b, s\rangle = 0$, $\delta = 2$ and $\alpha = 4$. For k: the 13056 pairs of Lemma 3.4 join the two colors.

The stated choice. For $a = 22133/25000$ and $b = 27161/25000$, exact rational arithmetic gives $f(a)^2 = 1054687500/4239869689$, $g(b) \approx 0.49875310$ and $k(a,b) \approx 0.49875283$, all at most $1246883/2500000$.

The lower bound. Let $\gamma = 0.4987528$, $a_1 = 177063/200000$ and $b_1 = 1086437/1000000$. Exact arithmetic gives $f(a_1) > \gamma$, $g(b_1) > \gamma$ and $k(a_1, b_1) > \gamma$. Suppose the largest cosine were at most γ. Since f decreases, $f(a) \leq \gamma < f(a_1)$ forces $a > a_1$; since g increases, $b < b_1$. Now $2k(a,b)$ is the cosine of the angle between $(\sqrt{6}, a)$ and $(\sqrt{6}, b)$, and for $a_1 < a \leq b < b_1$ this angle is smaller than the one for (a_1, b_1). So $k(a,b) > k(a_1,b_1) > \gamma$, a contradiction. □

With the stated a, b, all coordinates of $25000\,p(c) = 75000\,q(c)_H + (25000\lambda_c)\tau(c)$ are integers. Giving a to $h = -1$ and b to $h = 1$ instead also passes the exact all-pairs check. The set A_1 is not closed under complement, so we do not know whether the two assignments are isometric.

5.5. **Nonlinear motions that fix the body.** The slack in Proposition 5.2 survives a family of motions that are not isometries (see the remark after Proposition 5.3). Work with unit vectors in $s^\perp$ and fix an orthonormal basis of $s^\perp$ that contains e_i for all $i \in H$. In unit normalization the head coordinates of body points lie in $\{0, \pm 1/(2\sqrt{2}), \pm 1/\sqrt{2}\}$, so

$$P(z) = 128\, z\Big(z^2 - \frac{1}{8}\Big)\Big(z^2 - \frac{1}{2}\Big)$$

vanishes at every head coordinate of every body point. A *gate* is given by two basis vectors e_α, e_β, a head coordinate $\kappa \in H$ with $e_\kappa \notin \{e_\alpha, e_\beta\}$, and a real number θ. It maps a unit vector x to the vector obtained by replacing (x_α, x_β) with

$$\frac{\big((1-u^2)x_\alpha - 2u\, x_\beta,\ 2u\, x_\alpha + (1-u^2)x_\beta\big)}{1+u^2}, \qquad u = \theta P(x_\kappa).$$

This is a rotation of the (α, β) plane by the angle $2 \arctan u$, so norms are preserved. The coordinate x_κ does not change, so the gate with $-\theta$ is its inverse. A gate fixes every point with $P(x_\kappa) = 0$, in particular every body point.

Proposition 5.3. *Start from the two-color configuration with $a = 22133/25000$ and $b = 27161/25000$, normalized. Apply any finite sequence of gates $g_1, \dots, g_m$ with parameters satisfying $\sum_k |\theta_k| \le 3/32768$ (for example 24 gates with $|\theta_k| \le 2^{-18}$). The image is a kissing configuration of 8358 points in $s^\perp$ that contains the 7398 body points, and every cosine involving an added point is at most*

$$\frac{1246883}{2500000} + \frac{1}{1024} = \frac{39978381}{80000000} < \frac{1}{2}.$$

The idea is that each gate moves an added point by at most a constant times $|\theta_k|$, as long as the head coordinates stay small. The total motion is then at most $1/2048$, and this is less than the slack $\frac{1}{2} - \frac{1246883}{2500000}$ of the two-color configuration.

Proof. The body is fixed by every gate. Every head coordinate of an added unit point is $\pm 75000/N_c$ with $N_c = |25000\, p(c)|$. Since $N_c^2 > 16 \cdot 75000^2$, these coordinates have absolute value less than $1/4$.

For $|z| \le 1/3$ we have $|z^2 - 1/8| \le 1/8$ and $|z^2 - 1/2| \le 1/2$, so $|P(z)| \le 128 \cdot \frac{1}{3} \cdot \frac{1}{8} \cdot \frac{1}{2} = \frac{8}{3}$. A rotation by angle $2 \arctan u$ moves a vector of norm at most 1 by at most $2|u|/\sqrt{1+u^2} \le 2|u|$. Let $x^{(k)}$ be the image of an added point after k gates. We claim by induction on k that

$$|x^{(k)} - x^{(0)}| \le \frac{16}{3} \sum_{j \le k} |\theta_j| \le \frac{16}{3} \cdot \frac{3}{32768} = \frac{1}{2048}.$$

If this holds for k, then every head coordinate of $x^{(k)}$ has absolute value at most $\frac{1}{4} + \frac{1}{2048} < \frac{1}{3}$. So the next gate has $|u| \le \frac{8}{3}|\theta_{k+1}|$ and moves $x^{(k)}$ by at most $\frac{16}{3}|\theta_{k+1}|$, and the claim holds for $k+1$.

So every added point moves by at most $D = 1/2048$. An inner product between a body point and an added point changes by at most D, and one between two added points by at most $2D = 1/1024$. By Proposition 5.2 all these products start at most $1246883/2500000$. Body pairs are unchanged, and the body spans $s^\perp$. □

For rational θ_k the coordinates are algebraic. Every added point has $0 < |x_\kappa| < 1/4$, so $P(x_\kappa) \ne 0$, and its coordinates (x_α, x_β) are not both zero. So a single gate with $\theta \ne 0$ moves every added point and fixes the body. Since the body spans $s^\perp$, the only isometry of $s^\perp$ that fixes the body is the identity, so the gate does not act on the configuration as an isometry. The gates never add a point.

6. Which representatives are different

Two configurations that differ by an orthogonal map have the same number of contacts, the same multiset of cosines, and either both are 3-designs or neither is. Table 3 lists these invariants for the four explicit representatives. The frame invariant is $\operatorname{tr}(S^2)$ for $S = \sum_x xx^{\mathsf{T}}$; a 3-design has $\operatorname{tr}(S^2) = 8358^2/18 = 3\,880\,898$, and $\operatorname{tr}(S^2) \geq 8358^2/18$ in general.

Table 3. Invariants of the four representatives, computed exactly. The third column is the largest cosine strictly below 1/2; body pairs have cosine 1/2 or at most 1/4, since inner products of body vectors are even integers.

representative	contacts	largest cosine below $\frac{1}{2}$	$\operatorname{tr}(S^2)$
projection, $\lambda = 1$	1 367 430	$\sqrt{27/112} \approx 0.49099$	3 881 070.73 …
irregular rational	1 354 374	≈ 0.499999975	3 880 997.80 …
3-design, $\lambda^2 = 174/151$	1 367 430	≈ 0.49763	3 880 898
two-color	1 354 374	≈ 0.4987531	3 881 129.86 …

Proposition 6.1. *The projection, the irregular choice in the data, the 3-design and the two-color choice of Proposition 5.2 are pairwise non-isometric.*

Proof. Contacts separate the two pairs: the projection and the 3-design have 1 367 430 contacts, the other two have 1 354 374. Within the first pair, the 3-design is a 3-design and the projection is not, since its $\operatorname{tr}(S^2)$ exceeds $8358^2/18$. Within the second pair, the irregular choice has a pair with cosine in $(1246883/2500000,\ 1/2)$, namely one with exact squared cosine $121719347689/486877439356$, and the two-color choice has none by Proposition 5.2. □

We do not claim more. Whether the four signed normals of Remark 3.6 give isometric configurations, whether the two color assignments are isometric, and which members of the family of Theorem 4.1 are isometric to each other, we leave open. Kravatskiy's configuration [5] is not compared here.

7. Verification

By hand: Lemmas 2.2, 2.3, 3.1, 3.2 and 3.3, the first part of Lemma 2.1(3), Theorem 3.5, Theorem 4.1 given Lemma 3.4, Corollary 4.2 and Proposition 5.3 given Proposition 5.2. By computer, in exact arithmetic: the rest of Lemma 2.1, Lemma 3.4, the counts in Remark 3.6, the numerical comparisons in Propositions 5.1, 5.2 and 6.1, the irregular margin, the contact counts and the checks below.

All pairs. The projection, the irregular choice and the two-color choice are given as integer vectors in $\mathbb{R}^{19}$. Every vector satisfies $x_2 + x_{11} + x_{14} = 0$, so all lie in $s^\perp$, and they span $s^\perp$ because the body does. One program checks each file on all $\binom{8358}{2} = 34\,923\,903$ pairs, with no symmetry reduction and no floating point in any decision. It confirms that each stated squared norm is the sum of squares of the coordinates. For every pair x, y with $\langle x, y\rangle > 0$ it compares $4\langle x, y\rangle^2$ with $|x|^2|y|^2$ in 128-bit integers. Since the vectors lie in $s^\perp$ and inner products are those of $\mathbb{R}^{19}$, this is the 18-dimensional check. Table 4 gives the results. A copy with one coordinate changed is rejected.

A second exact program, written independently, checks the projection and the irregular choice on all pairs, confirms distinct directions, and computes the rank exactly: 18, with an 18×18 integer minor of determinant -524288 and the normal vector $e_2 + e_{11} + e_{14}$. An earlier

independent rebuild of the section from Ho's published coordinates gave the same set of vectors as the projection. A third computation checks every pair involving an added point of all three files in exact integer arithmetic, and checks the 3-design pairs exactly in $\mathbb{Q}(\beta)$, where body and added points have products $u + v\beta$ with integers u, v.

Table 4. The exact all-pairs check in dimension 18.

representative	points	pairs	violations	result
projection	8358	34 923 903	0	pass
irregular rational	8358	34 923 903	0	pass
two-color	8358	34 923 903	0	pass

8. Conclusion

The section of Ho's 19-dimensional configuration by $s^{\perp}$, with the 960 projected sign vectors, gives $\tau_{18} \geq 8358$, equal to Kravatskiy's repository value and 704 above the best published bound. The added points have a lot of freedom: each tail factor moves in an interval, the irregular choice frees every added point, and gates move them nonlinearly. All of these motions keep the number of points at 8358, and none of them has made room for an 8359th point.

Use of artificial intelligence

GPT-6-Astra Pro and Claude Opus 5.5 were used to find these constructions and draft this paper.

Data availability

The coordinates of the projection, the irregular choice and the two-color choice, and an exact checking program, accompany this paper.

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